Construction Chemicals

SUBMITTAL DOCUMENT

Waterbar

PVC Waterstop for Construction & Expansion Joints in Concrete



Watertight
Up to 50 m water head

Movement
Up to 10 mm

Profiles
Centerstop & Outerstop

Site Welding
Heat-welded joints

Project	
Main Contractor	
Consultant	
Submittal No. / Rev.	00
Date	Nov 2026

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METHOD STATEMENT

Waterbar — PVC Waterstop for Construction & Expansion Joints in Concrete

Project Name		**Document No.**	OMX-MS-WATERBAR
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
Product	Waterbar	**Status**	For Approval

1. Purpose

This method statement defines the procedure, sequence, resources, quality checks and safety measures for the supply and application of Waterbar on site. It ensures that the work is carried out safely, in line with the approved drawings and project specification, and achieves the performance stated in the O MIX Technical Data Sheet.

2. Scope of Work

This method statement covers the selection, positioning, fixing, heat welding, inspection and protection of Waterbar PVC waterstops in construction and expansion joints of cast-in-place concrete structures.

- Review of joint layout and selection of waterstop profiles.
- Cutting, heat welding and installation of junction pieces.
- Positioning and fixing to formwork and reinforcement.
- Inspection of the installation before concreting.
- Protection during concreting and preparation of the second pour.

3. Definitions

Term	Meaning
Centerstop	Waterstop placed in the centre of the concrete section.
Outerstop	Waterstop placed on the external face of the concrete, against the formwork.
Construction joint	Joint between two pours without designed movement.
Expansion joint	Joint designed to allow movement; requires a bulb profile.
Butt weld	Joint formed by melting the profile ends with a heated blade and pressing together.

4. Roles & Responsibilities

Role	Responsibility
Project Manager	Overall responsibility for planning, resources, approvals and coordination with the consultant and main contractor.

Role	Responsibility
Site Engineer	Supervises the work daily; confirms substrate readiness, climate conditions, mixing, consumption and sequence; keeps daily records.
QA/QC Inspector	Carries out inspections and tests at each stage, records results, raises inspection requests at hold points and signs the checklist.
HSE Officer	Conducts toolbox talks, checks PPE, ventilation, access and safe storage; stops work if unsafe conditions arise.
Foreman / Applicators	Carry out the work as per this method statement and report any deviation immediately.
Storekeeper	Receives, stores and issues materials in original packaging under the stated storage conditions; controls batch numbers.

5. Materials

Material	Description
Waterbar	PVC waterstop profiles as per joint schedule (Centerstop / Outerstop)
Junction pieces	Prefabricated T, L and cross pieces
Fixing materials	Tie wire, waterstop clips, split formwork

5.1 Key product data

Property	Typical Value
Material	Flexible PVC
Hardness	> Shore A 65
Tensile strength	≥ 12 MPa
Elongation at break	≥ 300 %
Brittleness temperature	< -38 °C
Water pressure resistance	up to 50 m head
Joint movement	up to 10 mm
Colours	Blue, black, yellow, grey

6. Tools & Equipment

- Thermostatic welding blade
- Cutting knife
- Mitre guide
- Tape measure
- Wire cutters
- Brushes
- Thermo-hygrometer and surface thermometer (dew point check)
- Moisture meter
- Measuring buckets and scale

- Masking tape and protective sheeting
- Lighting and extension cables

6.1 Personal protective equipment

- Gloves
- Safety glasses
- Helmet
- Safety shoes

7. Storage & Handling

- 12 months in original packaging. Store rolls flat in a dry, shaded place away from direct sunlight, heat and oil.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Store rolls and cartons flat, off the ground, protected from direct sunlight, heat, oil and mechanical damage.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

8. Pre-Start Requirements

- Material submittal, this method statement and the inspection plan approved by the consultant.
- Work area handed over clean, free of other trades and with safe access.
- Adequate lighting, power supply and ventilation available.
- Climate conditions checked and recorded before work starts.
- A trial area prepared and approved where required by the project.
- Toolbox talk held with the work team covering this method statement and the safety measures.

9. Surface Preparation

1. Review the joint layout drawings and confirm the profile at each joint.
2. Prepare split formwork or stop-ends that allow half of the waterstop to pass through.
3. Clean the waterstop of dirt, oil and mud.
4. Plan the positions of welds and junction pieces so that site welds are straight butt welds only.
5. Obtain QA/QC approval of the joint layout and profiles before installation.

10. Cutting & Heat Welding

1. Check the profile type, size and batch of the delivered rolls.
2. Cut the profile ends square using a sharp knife and mitre guide.
3. Heat the welding blade to the working temperature.
4. Press both ends against the blade until the PVC melts evenly across the full section.

5. Remove the blade and press the ends together, aligned, until cooled.
6. Inspect the weld: continuous bead, no gaps, voids or misalignment; bend test the joint.

11. Installation

1. Position the waterstop centrally in the joint (Centerstop) or on the formwork face (Outerstop).
2. Pass half of the width through the split formwork; keep the bulb exactly on the joint line for expansion joints.
3. Fix the edges to reinforcement with tie wire through the edge holes or clips at max. 300 mm centres.
4. Install junction pieces at intersections and weld to the straight runs.
5. Place and compact concrete carefully around the waterstop, avoiding displacement or folding.
6. Before the second pour, clean the exposed half and confirm its position.
7. Record the inspected length and welds per joint.

12. Curing & Protection

- Protect the exposed half of the waterstop from damage, UV and contamination until the second pour.
- Do not bend the exposed half repeatedly.
- Repair any damage by heat welding a new section before concreting.

13. Quality Control & Inspection

Check	Stage	Acceptance Criteria	By
Profile	Before installation	As per joint schedule	Site Engineer
Welds	Each weld	Continuous, no gaps, passes bend test	QA/QC
Position	Before concreting	Central, bulb on joint line, half in each pour	QA/QC
Fixing	Before concreting	Max. 300 mm centres both edges	QA/QC
After pour	After concreting	Not displaced; exposed half clean	Site Engineer

13.1 Inspection and test plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Profile, size, condition	Each delivery	Material inspection report
Welds	Visual and bend test	Each weld	Weld log
Installation	Position, fixing centres, alignment	Each joint	Inspection request (hold point)
Concreting	Position after pour	Each pour	Pour record
Second pour	Cleanliness of exposed half	Each joint	Inspection request

14. Defects & Remedies

Defect	Likely Cause	Remedy
Displaced or folded waterstop	Insufficient fixing; careless concreting	Fix at 300 mm centres; place concrete in layers
Weld gaps or voids	Wrong temperature; contaminated ends	Cut out and re-weld
Honeycombing around waterstop	Poor compaction	Repair concrete; vibrate carefully in next pours
Lapped instead of welded joints	Incorrect practice	Remove and replace with butt welds

15. Health, Safety & Environment

Hazard	Control Measures
Cuts from knives and tools	Cut-resistant gloves; sharp tools kept in holders
Work in excavations / at height	Shoring, approved access, harness where required
Hot-air welding equipment	Trained operators, inspected equipment, fire extinguisher nearby
Manual handling of rolls	Correct lifting technique; two-person lift for heavy rolls
Slips and trips	Housekeeping; barricades and signs

- Wear gloves and safety glasses when cutting and welding.
- Fumes from overheating PVC are irritant; avoid overheating and ensure ventilation.
- Follow site safety rules for work in excavations and on formwork.
- Burns from welding blade: cool with clean water and seek medical advice.
- Eyes: rinse with clean water.

15.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

16. Cleaning

- Clean the welding blade with a dry cloth while warm; remove concrete splashes from the waterstop with water before they set.

17. Limitations

- Do not lap, glue or nail through the waterstop.
- Bulb profiles are required at expansion joints.

- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Joint layout and profiles approved	☐ Yes ☐ No	
Split formwork prepared	☐ Yes ☐ No	
All site joints butt-welded	☐ Yes ☐ No	
Junction pieces installed	☐ Yes ☐ No	
Welds inspected and bend-tested	☐ Yes ☐ No	
Waterstop centred in joint	☐ Yes ☐ No	
Fixed at 300 mm centres	☐ Yes ☐ No	
Concrete compacted around profile	☐ Yes ☐ No	
Exposed half protected	☐ Yes ☐ No	
Second pour joint cleaned	☐ Yes ☐ No	

Inspected by	Signature	Date

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Technical Data Sheet

Waterbar — Submittal Document

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Product Datasheet
Edition: 01 | Date: Nov 2026
Waterproofing

Waterbar

PVC Waterstop for Construction & Expansion Joints in Concrete



Product Description

Waterbar is a range of flexible PVC waterstops cast into concrete to seal construction and expansion joints against water penetration. It is supplied as centre-placed (Centerstop) and externally-placed (Outerstop) profiles, with or without a centre bulb, and is joined on site by heat welding. Prefabricated T, L and cross junctions are available.

Uses

- Water tanks and reservoirs.
- Dams, channels and culverts.
- Swimming pools.
- Basements and foundations.
- Tunnels and retaining walls.

Advantages

- Reliable watertight seal in concrete joints.
- Accommodates joint movement up to 10 mm.
- Resists water pressure up to 50 m head.
- Resistant to water, soil and mild chemicals.
- Easily heat-welded on site; prefabricated junctions.

Application

1. **Positioning**
   Fix the waterstop centrally in the joint (Centerstop) or on the formwork face (Outerstop), with half the width in each pour.
2. **Fixing & Jointing**
   Secure to reinforcement with tie wire through the edge holes or clips; heat-weld all joints and use prefabricated junctions.
3. **Concreting**
   Place and compact concrete carefully around the waterstop without displacing or folding it.

Consumption

1 m per 1 m of joint
plus overlap and welding allowance
Use prefabricated T, L and cross pieces at junctions.

Packaging

Rolls of 15 m, 20 m, 25 m or 30 m depending on profile.

Storage & Shelf Life

12 months in original packaging. Store rolls flat in a dry, shaded place away from direct sunlight, heat and oil.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Material	Flexible PVC	—
Hardness	> Shore A 65	—
Tensile strength	≥ 12 MPa	—
Elongation at break	≥ 300 %	—
Brittleness temperature	< -38 °C	—
Water pressure resistance	up to 50 m head	—
Joint movement	up to 10 mm	—
Colours	Blue, black, yellow, grey	—
Shelf life	12 months	—

Consumption

Profile	Head	Width / Thickness	Roll
V-20 Centerstop — construction joint	10 m	20 cm / 3 mm	30 m
V-24 Centerstop — construction joint	15 m	24 cm / 3 mm	30 m
V-30 Centerstop — construction joint	25 m	30 cm / 4 mm	30 m
O-20 Centerstop with bulb — expansion	10 m	20 cm / 3.5 mm	15 m
O-24 Centerstop with bulb — expansion	25 m	24 cm / 3.5 mm	15 m
AR-25 / DR-25 Outerstop	—	26 cm / 5 mm	—

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Select bulb profiles for expansion joints and plain/ribbed profiles for construction joints.
- Embed half of the waterstop width in each concrete pour.
- Concrete cover and member thickness must allow full embedment of the profile.
- Heat-weld joints with a thermostatically controlled welding blade; do not lap or glue.
- Keep the waterstop clean and free of concrete laitance, oil and mud before the second pour.

Application Notes

- Fix at maximum 300 mm centres along both edges to prevent displacement.
- Use prefabricated T, L and cross junctions with 25 – 30 cm legs.
- Outerstop profiles: cut the notch of the bulb type to keep the barrier continuous at expansion joints.
- Inspect all welds visually before concreting.
- Protect exposed waterstop from damage and UV until the second pour.

Health & Safety

- Wear gloves and safety glasses when cutting and welding.
- Fumes from overheating PVC are irritant; avoid overheating and ensure ventilation.
- Follow site safety rules for work in excavations and on formwork.

First Aid & Cleaning

- Burns from welding blade: cool with clean water and seek medical advice.
- Eyes: rinse with clean water.
- Inhalation of welding fumes: move to fresh air.
- Clean the welding blade with a dry cloth while warm; remove concrete splashes from the waterstop with water before they set.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.

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Test Reports

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To be attached: laboratory test reports supporting the values in the Technical Data Sheet.

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Approvals & Certificates

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To be attached: approvals and certificates, where available.